# Remagnetization Problems With the Paleomagnetism of the Middle Silurian Rose Hill Formation of the Central Appalachians

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Ninety-nine red bed samples were collected from 15 sites of the Rose Hill Formation in the central Appalachians of West Virginia and Virginia. These hematite-bearing samples were demagnetized with thermal and chemical techniques which gave mutually consistent results. Most samples had reversed polarity directions which in some cases are undoubtedly of secondary (late Paleozoic) origin. Other reversed directions may be of Middle Silurian age, but this cannot be unambiguously demonstrated, since corrections for the tilt of the strata do not always provide a conclusive fold test. On the other hand, 17 samples showed normal polarity directions with northerly declinations. Some of these samples contained only one stable component of magnetization, whereas others reached a stable endpoint only after dominant reversed polarity components of magnetization had been removed by thermal demagnetization. Five of these samples did not reach a stable endpoint. The northerly directed components of normal polarity show a marked improvement in grouping after unfolding, indicating that these directions were acquired before the Appalachian folding. Their mean direction of magnetization, with declination 344° and inclination −42° ($k$ = 10.4, $\alpha_{95}$ = 14°, and $N$ = 12), yields a pole at 26°N, 116°E, which is close to published poles for North America of Silurian and Ordovician age. Although this pole should not be considered as firmly established, we consider it to be more reliable than pole positions obtained (through whatever selection criteria) from the reversed polarity samples. Apparently, the directions and blocking temperature spectra of the secondary and reversed polarity original magnetizations are too similar to allow their separation or recognition in thermal demagnetizations.

## INTRODUCTION

In a companion paper [*Van der Voo and French,* 1977] we reported on the paleomagnetism of the Late Ordovician Juniata Formation from the central Appalachians. It was found that despite the presence of a strong secondary magnetization, directions of characteristic magnetization could be fairly well determined. Measurements and demagnetizations on samples from the Rose Hill Formation had been performed earlier, but not until the Juniata results became available were we able to sort out the somewhat confusing results of the Rose Hill Formation, first reported by *Grubbs et al.* [1975] and *French* [1976].

The Rose Hill Formation is exposed in the valley and ridge province of the Appalachians from Alabama to northern Pennsylvania. It varies from a friable red gray shale in the north to a hematitic sandstone in our sampling area in Virginia and West Virginia [*Dennison,* 1970*a*, *b*]. Rose Hill is a formation of the Clinton Group of Niagaran, or Middle Silurian, age. Sampling localities are shown in Figure 1. The area has been extensively folded during the late Paleozoic, and in the south there are thrust faults on the western flanks of several folds.

Previous paleomagnetic results (but without demagnetization studies) for the Rose Hill Formation were published by *Graham* [1949], who sampled in Maryland. From 35 samples a pole at 20°N, 136°E has been calculated [*Cox and Doell,* 1960]. *Howell et al.* [1958] reported that seven samples of the facies equivalent Clinton iron ore in Alabama give a pole at 34°N, 139°E. Again, no demagnetizations were carried out, so little reliability can be attached to these poles, despite the fact that Graham's results passed the fold test. It should be noted that a small amount of magnetic overprinting may not be revealed decisively by the fold test when the folds are tight or isoclinal or when the direction of the magnetic overprinting is close to the fold axis.

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## PALEOMAGNETIC RESULTS

The directions of the natural remanent magnetizations have a preferred grouping in the SE quadrant near the horizontal (Figure 2*a*). Most intensities of magnetization, measured on a Schonstedt spinner magnetometer and a ScT cryogenic magnetometer, ranged between $10^{-2}$ A m$^{-1}$ ($10^{-5}$ emu/cm$^3$) and $5 \times 10^{-2}$ A m$^{-1}$. A majority of the directions, obtained through chemical and thermal demagnetization, still have southeasterly directions (Figure 2*b*). Their site means are shown in Figures 2*c* and 2*d*, plotted with respect to the present-day horizontal and paleohorizontal, respectively. Upon correction for the dip of the strata, most site means show a somewhat reduced scatter (Figure 2*d*). However, four of the site means diverge (Figure 2, triangles), an indication that at least for these sites the magnetizations were acquired after the folding.

Ninety-four samples were thermally demagnetized. Eighty of these yielded stable directions of magnetization, distinct from the local present-day field direction. Three thermal demagnetization diagrams [after *Zijderveld,* 1967] are shown in Figure 3. Typically, a small component of viscous remanent magnetization is removed below 400°C. The remaining magnetization then decays to the origin, and there is no evidence for further multivectorial behavior. Five additional samples, as well as several companion specimens of thermally treated cores, were chemically demagnetized [*French,* 1976]. These gave essentially the same results as the thermal demagnetizations. Figure 3 shows a chemical demagnetization for sample FVB 111 taken less than 3 m away from sample FVB 113 of Figure 3. Other chemically demagnetized samples showed a similar behavior, a small viscous component being removed in the first 15–150 hours of leaching, followed by decay to the origin.

A linear decay of the magnetization vector toward the origin of the Zijderveld diagrams, as in Figure 3, is usually interpreted as indicating that only one component of magnetization is being removed. Equivalently, this is called the stable endpoint [*McElhinny,* 1973, p. 93]. In theory, though, it is conceiv-

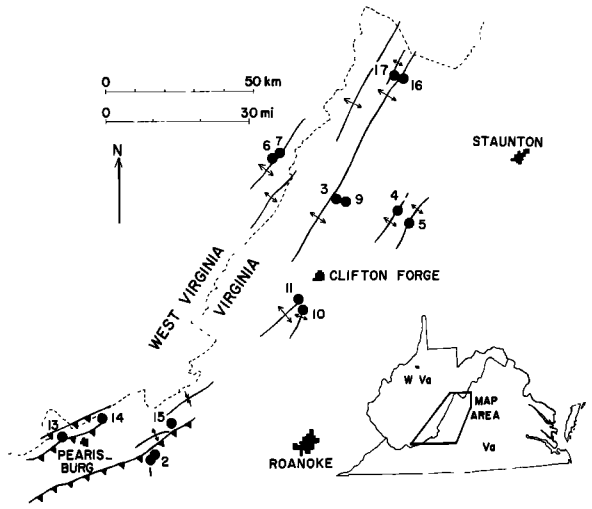



Fig. 1. Map of the Virginia-West Virginia border area showing the site locations of the Rose Hill samples.

able that two components of magnetization have exactly the same blocking temperature spectra. They would then be eliminated at the same time during thermal demagnetization, and this would result in a straight decay toward the origin. In practice, such situations are rarely found; however, they are difficult to recognize, since no magnetic analysis exists with which to document this situation. In order to arrive at an interpretation one must rely on field observations or paleomagnetic expectations, and therefore such interpretations are often based on assumptions rather than observations. In view of our suspicion that both secondary and original magnetizations may be present in some samples of the Rose Hill Formation, as was found also for most samples of the Juniata Formation [*Van der Voo and French*, 1977], the 'cleaned' directions of the Rose Hill samples such as those of Figure 2 cannot a priori be regarded as representing Silurian magnetizations.

In fact, one can argue that several arguments exist that the southerly directions cannot be used to give a reliable Silurian pole position. As was mentioned above, at least four sites clearly fail the fold test, and of the remaining sites, at least three (5, 9, and 17 in Figure 2) are barely conclusive. Although almost all of the reversed directions exhibited stable endpoints during demagnetization, many of these directions are near the Late Carboniferous and Permian directions as expected for this area [*Van der Voo and French*, 1974]. This makes the directions suspect in that they may not represent the original magnetizations but rather a complete remagnetization [*Creer*, 1968; *McElhinny*, 1973; *McElhinny and Opdyke*, 1973]. Which of the sites represent original (Silurian) magnetizations and which represent secondary magnetizations is impossible to decide without circular reasoning. In view of all this we have decided not to publish a pole position for the southerly directions, the details of which can be found in the paper by *French* [1976].

A few samples showed northerly directions, however, presumably of normal polarity (Figure 4*b*). It is unlikely that these directions represent secondary magnetizations. They show a significant improvement in clustering upon unfolding,

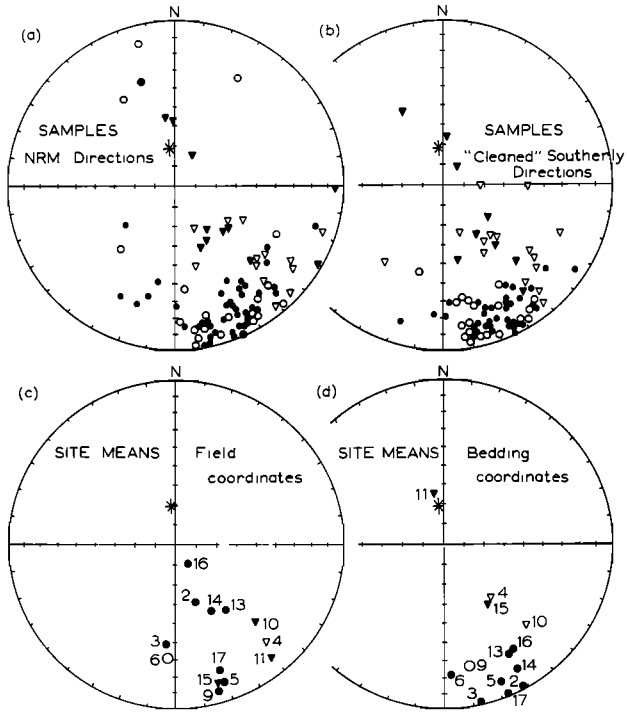



Fig. 2. Equal area projections of directions of magnetization. Open (solid) symbols represent projections on the upper (lower) hemisphere. The asterisk is the present-day geomagnetic field direction. Triangles represent directions of magnetization which show a negative fold test [*Graham*, 1949]. (*a*) Directions of the natural remanent magnetizations (NRM) of all samples, after correction for the tilt of the strata. (*b*) Directions obtained through demagnetization analysis, after correction for the tilt of the strata. Only magnetizations which had southerly declinations before or after tilt correction have been plotted. (*c*) Site mean directions plotted before tilt correction. (*d*) Site mean directions plotted after tilt correction.

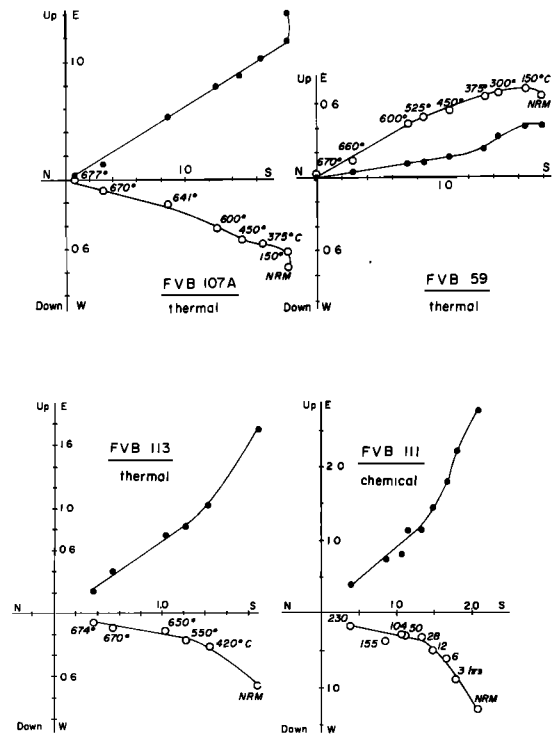



Fig. 3. Orthogonal demagnetization [*Zijderveld*, 1967] of the successive endpoints of the magnetization vector during progressive demagnetization, after tilt correction. Demagnetization temperatures (in degrees Centigrade) and cumulative leaching times (in hours) are indicated. Solid circles represent projections on the horizontal plane, and open circles those on the north-south vertical plane. Units indicated for the axes are in $10^{-2}$ A m$^{-1}$ ($10^{-5}$ emu/cm$^3$).

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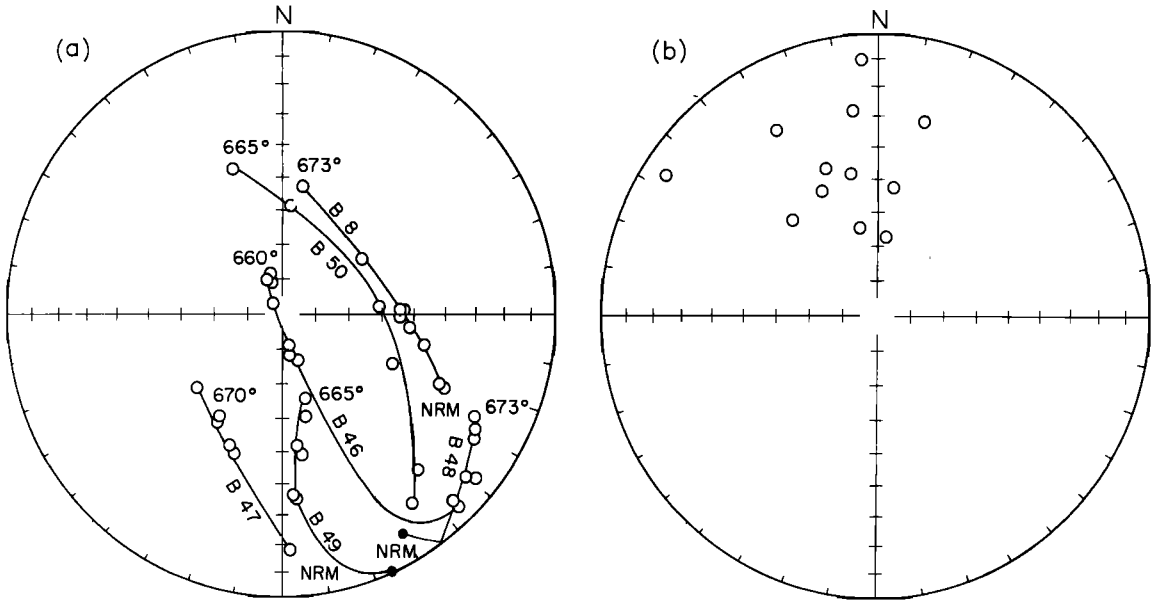



Fig. 4. (*a*) Equal area projection of the paths of successive positions of the endpoints of the magnetization vector during thermal demagnetization, after tilt correction. Open (solid) circles represent projections on the upper (lower) hemisphere. (*b*) Equal area projection of the characteristic directions of magnetizations with northerly declinations, after tilt correction.

$k$ increasing from 1.5 to 10.4 and $\alpha_{95}$ decreasing from 55° to 14° ($N$ = 12 samples), an indication that the magnetizations predate the folding. They are of normal polarity, whereas during the most likely period of remagnetization the field was predominantly reversed. Several samples showed the northerly directions only after southerly (secondary) magnetizations had been removed. As was the case with the Juniata samples, some samples show a progressive shift in direction (Figure 4*a*) without reaching a stable endpoint at high temperatures. However, other samples reached a stable endpoint at lower temperatures, and two of these are plotted in the Zijderveld diagrams of Figure 5. The mean direction ($D$ = 344.1°, $I$ = −42.0°) yields a pole position at 26°N, 116°E, which is very close to the Ordovician and Silurian poles for North America listed by *Van der Voo and French* [1977], as well as to the pole (21°N, 105°E) for the contemporaneous Castanea Formation [*McElhinny and Opdyke*, 1973].

In view of the limited number of samples we do not regard

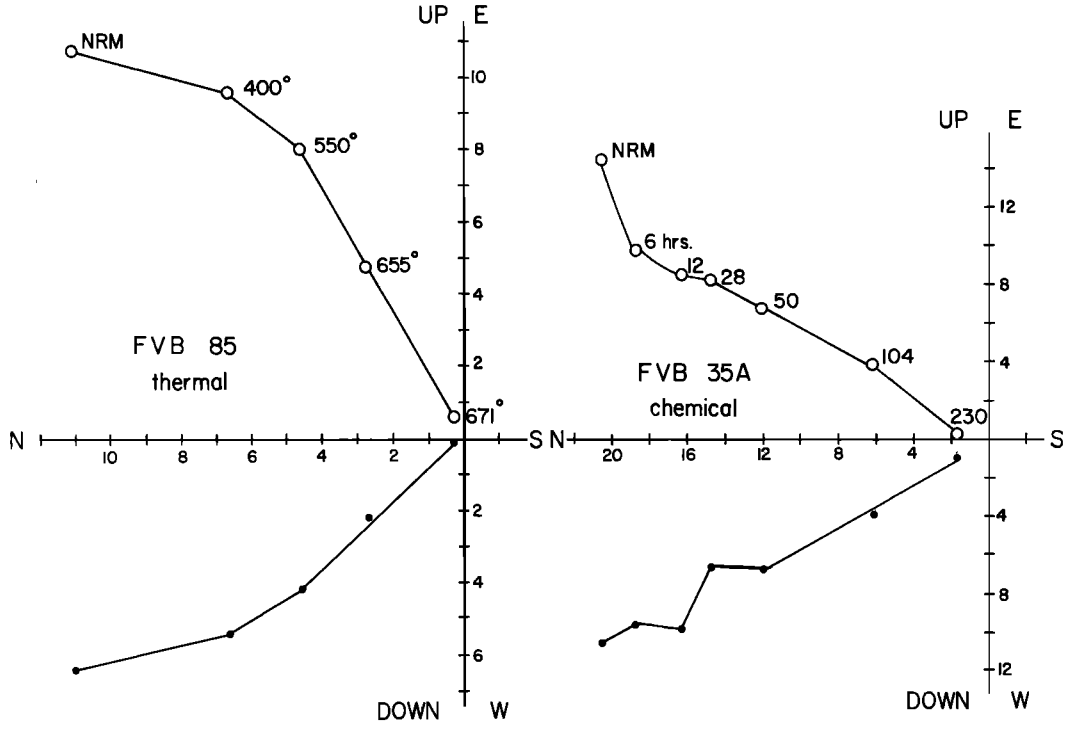



Fig. 5. Orthogonal demagnetization projections of two samples with northerly declinations, after correction for the tilt of the strata. Symbols are the same as those used in Figure 3; the units indicated for the axes are in $10^{-3}$ A m$^{-1}$.

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this pole position as firmly established. It may be worthwhile to investigate the facies equivalent formations in the southern Appalachians or in Maryland and Pennsylvania to obtain confirming results. In particular, it should be of interest to carry out demagnetization studies on samples from *Graham*'s [1949] sampling sites at Pinto, Maryland, to see whether partial or complete remagnetization occurred either before or after the folding.

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